Enterprise Systems Architecture

Computing and Mathematics

May, 2026

Ollscoil Teicneolaíochta an Oirdheiscirt
South East Technological University

Enterprise Systems Architecture (A14134)

Short Title:	Enterprise Systems Arch.		
Faculty	Science and Computing		
Department	Computing and Mathematics		
Cluster	Information Systems and Modelling		
Author	LDOYLE		
Credits	5	**Level:**	Advanced

Description of Module / Aims

This module aims to provide the student with an understanding of an architectural approach to development of information systems. The module introduces the constituent parts of architecture and their interrelationship. The student is introduced to the use of architectural frameworks and to the use of architecture approaches for different contexts within the organisation.

Programmes

`COMP-0610`	BSc (Hons) in Software Systems Development (`WD_KDEVP_B`)	4 / 7 / M
`COMP-0610`	BSc (Hons) in Software Systems Development (`WD_KCSDV_B`)	4 / 1 / M
`COMP-0610`	BSc (Hons) in Software Systems Practice (`WD_KSOFP_B`)	1 / 1 / M

Indicative Content

- Architectural Structure, Components, and Interactions
- Architectural Frameworks
- Enterprise Architectures
- Data Architectures
- Information System Architectures
- Technology Architecture
- Enterprise Modelling
- Enterprise Application Architectures and Integration
- Process Management and Service Architectures
- Model Driven Architectures
- Architecture Development Methods and Capabilities

Learning Outcomes

On successful completion of this module, a student will be able to:

1. Determine the role of architectures in the development of enterprise systems.
2. Construct architectures for business, data, technology, and enterprise applications.
3. Design a service architecture to support a set of business processes.
4. Evaluate the role of model driven architectures in enterprise application development.
5. Compare architecture development method.

Learning and Teaching Methods

- This module will be presented by a combination of lectures and computer based practicals.
- The lectures will be used to introduce new topics and their related concepts. Lectures will be supplemented by participative case studies and independent reading on the issues covered in the lecture material.

- The practical element is intended to provide the student with the skills needed to use and understand technologies available to assist in enterprise architecture design.

Assessment Methods

	Weighing	Outcomes Assessed
Final Written Examination	50%	1,3,4,5
Continuous Assessment	50%	
Assignment	50%	2,3

Assessment Criteria

<40%: Unable to interpret and describe key concepts of enterprise systems architecture.
40%-49%: Be able to interpret and describe key concepts of enterprise systems architecture.
50%-59%: Ability to discuss key concepts of enterprise systems architecture and ability to discover and integrate related knowledge in other knowledge domains.
60%-69%: Be able to solve problems within enterprise systems architecture by experimenting with the appropriate skills and tools.
70%-100%: All the above to an excellent level. Be able to analyse and design solutions to a high standard for a range of both complex and unforeseen problems through the use and modification of appropriate skills and tools.

Learning Modes

Learning Mode	F/T Hours	P/T Hours
Lecture	36	
Practical	12	
Independent Learning	87	

Essential Material

- Desfray, P. and G. Raymond. _Modelling Enterprise Architecture with TOGAF: A Practical Guide using UML and BPMN_. Waltham, MA, USA: Morgan Kaufmann/OMG Press, 2014.

Supplementary Material

- Bass, L. and P. Clements. _Software Architecture in Practice_. 3rd.. Upper Saddle River, NJ, USA: Pearson, 2013.
- Bernard, S.A. _An Introduction to Enterprise Architecture_. Bloomington, IN, USA: AuthorHouse, 2012.
- Fowler, M. _Patterns of Enterprise Application Architecture_. Upper Saddle River, NJ, USA: Pearson, 2002.
- Newman, S. _Building Microservices: Designing Fine Grained Systems_. Sebastapol, CA, USA.: O'Reilly, 2015.
- Rosenfeld, L., P. Morville and J. Arango. _Information_ _Architecture: For the Web and Beyond_. Sebastapol, CA, USA: O'Reilly, 2015.
- Russell-Rose, T. and T. Tate. _Designing the Search Experience: The Information Architecture of Discovery_. Waltham, MA, USA: Morgan-Kaufmann, 2013.

Requested Resources

- Room Type: Computer Lab
- Lecture Room: Loose Seated